

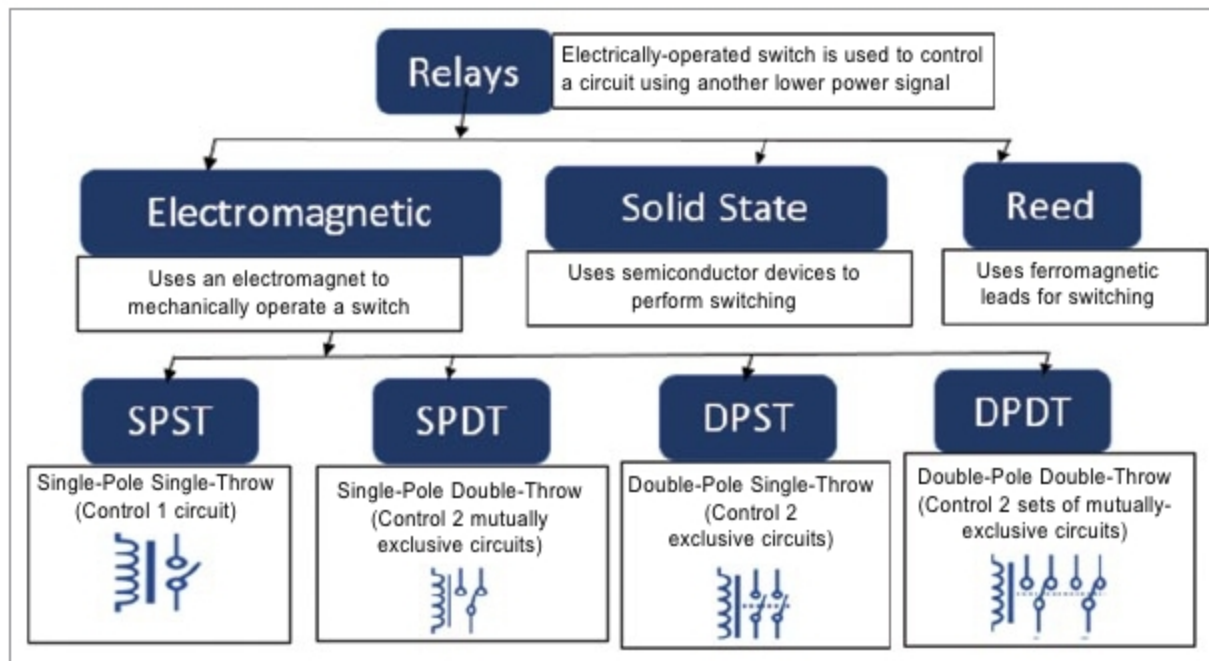


Fig. 2: Basic classification of electromagnetic relays

Reed relays use ferromagnetic material leads actuated by a magnetic field. The table shows the differences between these relays.

Electromagnetic relays, though bulky and slow, are extremely versatile and cost-effective. Fig. 2 summarises single-pole single-throw (SPST), single-pole double-throw (SPDT), double-pole single-throw (DPST) and

double-pole double-throw (DPDT) as the various categorises of electromagnetic relays based on the number of throws and poles.

Fig. 3 uses an SPDT electromagnetic relay to explain throw, pole, normally-open (NO) and normally-closed (NC) concepts. The number of external circuits that can be controlled constitutes poles, and the number of

connecting nodes constitutes throws.

Throw happens between NC and NO. The point to which the lever is connected in its resting position (with no excitation signal) is NC and not connected is NO. With no excitation signal, the circuit (pole) along with NC forms a closed external circuit, and with NO an open external circuit.

Now, let us talk about using above-mentioned electromagnetic relays for achieving different circuit operations through four applications. You can improvise and innovate based on your application's requirement. This is possible because of the versatile nature of electromagnetic relays, as these fit into every circuit. Arduino Uno is a very famous, cost-effective and simple development board.

Application 1: Smart control of traffic signals (using SPST relay). Controlling of traffic signals can be done using data acquired from vibration sensors on crossroads. SPST relays act as simple switches for controlling red-yellow-green signals, which work on higher power supply.

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